INSTITUT FÜR SOLARTECHNIK

OST Ostschweizer Fachhochschule

Lorenz model fitting for heat pump komotiniLorenz

Types 4050, 4051, 4052 (heating mode)

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2026/10/08 at: 17:06:23 h

Table 1: Fitted parameters of the Lorenz heat pump model in heating mode.

Parameter	Description		
η model	*saturation*	eta = a·(1-exp(-L/b))	
etaA	*Lorenz efficiency coefficient*	0.71327	fitted
etaB	*Lorenz efficiency coefficient*	150.73	fitted
η range	*Lorenz efficiency in the data*	0.0454 - 0.3616	
$\Delta T_{\text{app,evap}}$	*Approach temperature evaporator*	8.0000 [K]	fixed
$\Delta T_{\text{app,cond}}$	*Approach temperature condenser*	8.0000 [K]	fixed
RMS_{COP}	$2.24e-02$		
$RMS_{W_{comp}}$	$1.86e+00$		

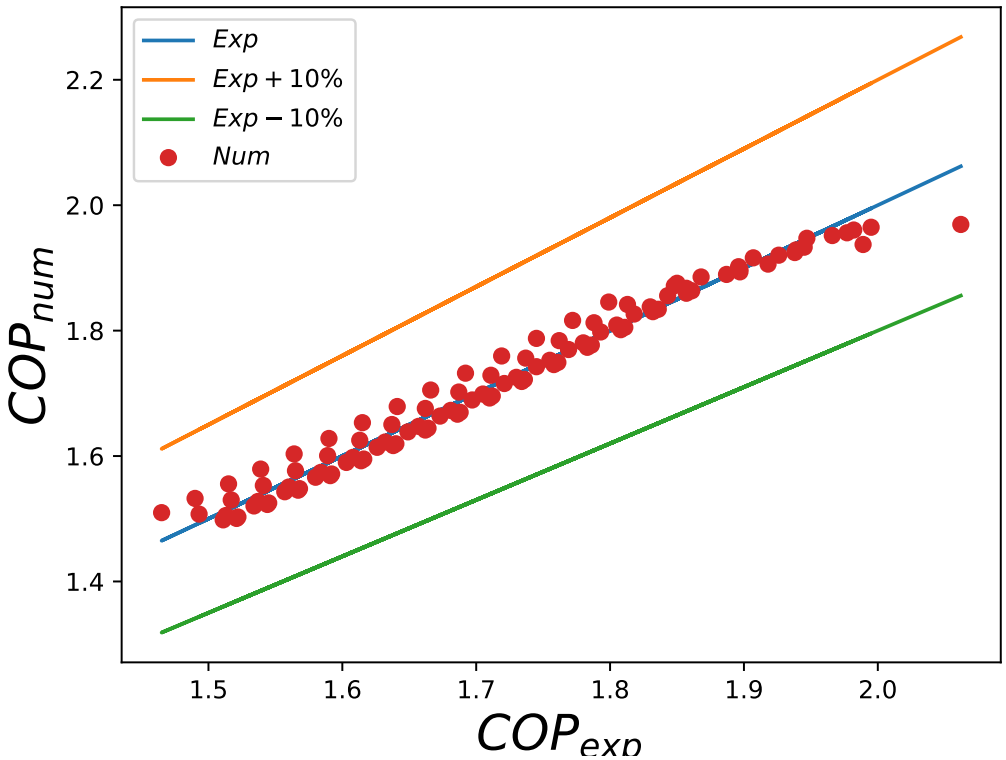


Figure 1: COP differences between experiments and fitted data

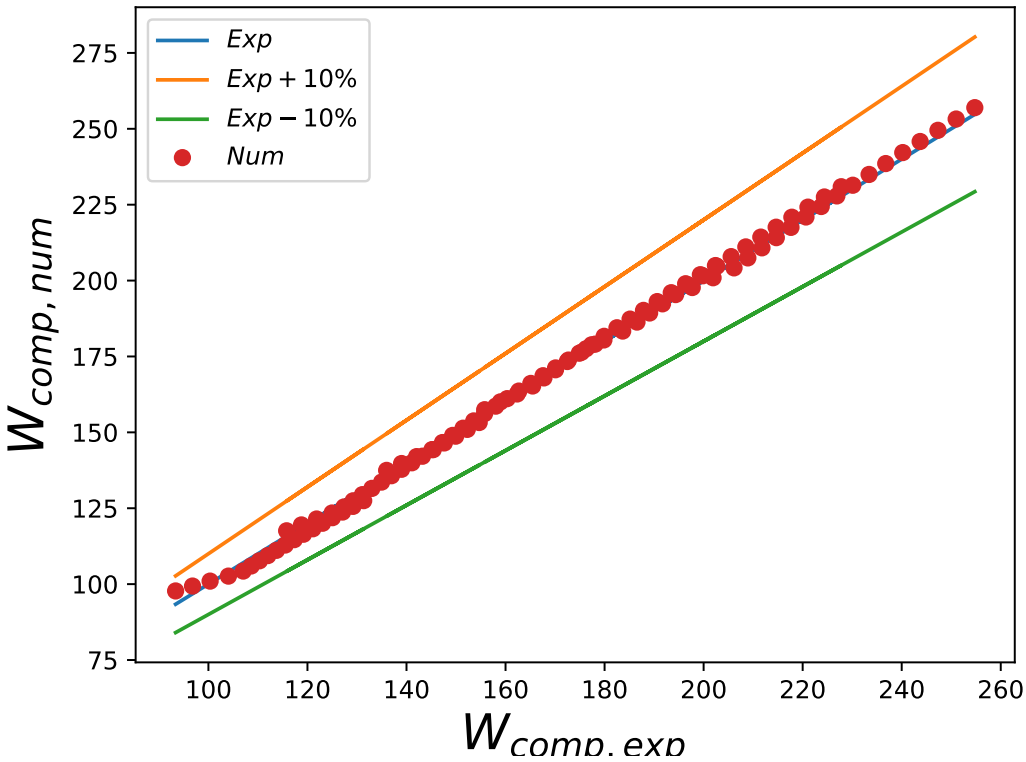


Figure 2: W_{comp} differences between experiments and fitted data

Table 2: Differences between experiments and fitted data for the heat pump. error $= 100 \cdot |\frac{X_{\text{exp}} - X_{\text{num}}}{X_{\text{exp}}}|$.

$T_{\text{evap,in}}$ °C	$T_{\text{evap,out}}$ °C	$T_{\text{cond,in}}$ °C	$T_{\text{cond,out}}$ °C	COP [-]	COP_{exp} [-]	error [%]	W_{comp} [kW]	$W_{\text{comp,exp}}$ [kW]	error [%]
80.00	60.00	136.76	185.10	1.498	1.511	0.8	256.97	254.76	0.9
85.00	65.00	136.76	185.10	1.520	1.534	0.9	253.22	251.00	0.9
90.00	70.00	136.76	185.10	1.543	1.557	0.9	249.50	247.32	0.9
95.00	75.00	136.76	185.10	1.566	1.580	0.9	245.82	243.72	0.9
100.00	80.00	136.76	185.10	1.590	1.603	0.8	242.17	240.21	0.8
105.00	85.00	136.76	185.10	1.614	1.626	0.7	238.55	236.77	0.7
110.00	90.00	136.76	185.10	1.639	1.649	0.6	234.96	233.41	0.7
115.00	95.00	136.76	185.10	1.664	1.673	0.6	231.41	230.12	0.6
120.00	100.00	136.76	185.10	1.689	1.697	0.4	227.89	226.90	0.4
125.00	105.00	136.76	185.10	1.716	1.721	0.3	224.40	223.75	0.3
130.00	110.00	136.76	185.10	1.742	1.745	0.1	220.95	220.66	0.1
135.00	115.00	136.76	185.10	1.770	1.769	0.0	217.53	217.64	0.1
140.00	120.00	136.76	185.10	1.798	1.793	0.3	214.14	214.68	0.2
145.00	125.00	136.76	185.10	1.826	1.818	0.5	210.79	211.78	0.5
150.00	130.00	136.76	185.10	1.856	1.843	0.7	207.47	208.93	0.7
155.00	135.00	136.76	185.10	1.886	1.868	0.9	204.18	206.14	0.9
160.00	140.00	136.76	185.10	1.916	1.907	0.5	200.93	201.91	0.5
165.00	145.00	136.76	185.10	1.947	1.947	0.0	197.71	197.71	0.0
80.00	60.00	136.76	180.35	1.501	1.521	1.3	230.92	227.79	1.4
85.00	65.00	136.76	180.35	1.523	1.544	1.4	227.54	224.39	1.4
90.00	70.00	136.76	180.35	1.546	1.567	1.4	224.19	221.06	1.4
95.00	75.00	136.76	180.35	1.569	1.591	1.4	220.87	217.82	1.4
100.00	80.00	136.76	180.35	1.593	1.614	1.3	217.58	214.65	1.4
105.00	85.00	136.76	180.35	1.617	1.638	1.3	214.32	211.55	1.3
110.00	90.00	136.76	180.35	1.641	1.662	1.2	211.09	208.52	1.2
115.00	95.00	136.76	180.35	1.667	1.686	1.1	207.89	205.56	1.1
120.00	100.00	136.76	180.35	1.693	1.710	1.0	204.72	202.67	1.0
125.00	105.00	136.76	180.35	1.719	1.734	0.9	201.58	199.84	0.9
130.00	110.00	136.76	180.35	1.746	1.758	0.7	198.47	197.06	0.7
135.00	115.00	136.76	180.35	1.773	1.783	0.5	195.39	194.35	0.5
140.00	120.00	136.76	180.35	1.802	1.808	0.4	192.34	191.70	0.3
145.00	125.00	136.76	180.35	1.830	1.832	0.1	189.32	189.09	0.1
150.00	130.00	136.76	180.35	1.860	1.857	0.1	186.33	186.54	0.1
155.00	135.00	136.76	180.35	1.890	1.887	0.1	183.37	183.66	0.2
160.00	140.00	136.76	180.35	1.920	1.926	0.3	180.44	179.86	0.3
165.00	145.00	136.76	180.35	1.952	1.966	0.7	177.54	176.27	0.7
80.00	60.00	136.76	175.57	1.503	1.522	1.3	204.95	202.39	1.3
85.00	65.00	136.76	175.57	1.525	1.545	1.3	201.94	199.34	1.3
90.00	70.00	136.76	175.57	1.548	1.568	1.3	198.96	196.37	1.3
95.00	75.00	136.76	175.57	1.571	1.592	1.3	196.01	193.46	1.3
100.00	80.00	136.76	175.57	1.595	1.616	1.3	193.08	190.63	1.3
105.00	85.00	136.76	175.57	1.620	1.640	1.2	190.18	187.86	1.2
110.00	90.00	136.76	175.57	1.644	1.664	1.2	187.30	185.15	1.2
115.00	95.00	136.76	175.57	1.670	1.688	1.1	184.46	182.50	1.1
120.00	100.00	136.76	175.57	1.696	1.712	1.0	181.63	179.91	1.0
125.00	105.00	136.76	175.57	1.722	1.736	0.8	178.84	177.38	0.8
130.00	110.00	136.76	175.57	1.749	1.761	0.7	176.07	174.89	0.7
135.00	115.00	136.76	175.57	1.777	1.786	0.5	173.33	172.47	0.5
140.00	120.00	136.76	175.57	1.805	1.811	0.3	170.62	170.09	0.3
145.00	125.00	136.76	175.57	1.834	1.836	0.1	167.93	167.75	0.1